

Digital Marketing Analytics

Channel Performance, ROAS & Customer Segment Analysis — Vaayu Wellness (D2C Brand)

Prepared by **Anupriya Bose** | Portfolio Project | Tools: Python (pandas, numpy, matplotlib, seaborn)

Vaayu Wellness is a fictional Indian D2C brand selling wellness teas, supplements, and skincare online. This project analyzes 12 months of daily marketing campaign data across 5 channels — Google Ads, Meta Ads, Influencer/Instagram, Email Marketing, and SEO/Organic — to answer the question every marketing team eventually has to answer with numbers, not opinions: **which channels are actually worth the money, and where should next quarter's budget go?**

The dataset is synthetic but built to mirror real Indian D2C e-commerce benchmarks — including seasonal demand spikes around Diwali and the Republic Day/New Year sale — so the analysis, numbers, and recommendations read the way a genuine quarterly marketing report would.

Headline Numbers (12 Months)



Why This Project

This comes from a digital marketing and content background — SEO writing, social media management, and e-commerce operations internships — built specifically to demonstrate the ability to back up marketing decisions with data, not just execute campaigns. It simulates exactly what a Marketing Analyst, Growth Analyst, or Digital Marketing Executive role expects: pull channel data, compute the right KPIs, find the story in the numbers, and turn it into a recommendation leadership can act on.

Contents

1. Methodology & KPI Definitions
2. Channel Performance Summary
3. Visual Analysis — Spend vs Revenue
4. Visual Analysis — ROAS by Channel
5. Visual Analysis — Monthly Seasonality Trend
6. Visual Analysis — Channel Efficiency Map
7. Customer Segment Analysis
8. Marketing Funnel Overview
9. Key Findings & Business Recommendations
10. How This Project Was Built

1. Methodology & KPI Definitions

The dataset contains daily records of impressions, clicks, conversions, spend, and revenue per channel, plus customer segment tags (city tier, age group), across 365 days (1,825 total rows: 5 channels x 365 days). Seasonality is modeled explicitly for the Diwali period (mid-October to mid-November), the Republic Day/New Year sale (late December to January 26), and a May summer sale — since ignoring festive demand spikes is one of the most common blind spots in real marketing reporting.

Metric	Formula	Business Question It Answers
CTR	Clicks / Impressions	Is the ad creative or targeting relevant?
CVR	Conversions / Clicks	Does the landing page or offer convert traffic?
CPC	Spend / Clicks	How expensive is a single click on this channel?
CAC	Spend / Conversions	How much do we pay to acquire ONE customer?
ROAS	Revenue / Spend	For every Rs 1 spent, how much revenue comes back?

Tools used: Python (pandas, numpy for KPI calculations; matplotlib and seaborn for visualization). This stack was chosen deliberately because it mirrors the tools used in real analyst roles and keeps every transformation transparent and auditable rather than hidden behind a BI tool.

2. Channel Performance Summary

The table below is sorted by total revenue, but the more important column for decision-making is **ROAS** — a channel can top the revenue chart simply because it received the largest budget, which does not necessarily mean it is the most efficient use of the next rupee spent.

Channel	Spend (Rs)	Revenue (Rs)	ROAS	CAC (Rs)	CVR
SEO / Organic	5.7 L	73.0 L	12.77x	68	4.68%
Google Ads (Search)	18.2 L	62.7 L	3.45x	255	5.43%

Meta Ads (FB/Insta)	16.5 L	54.4 L	3.30x	260	3.41%
Email Marketing	1.3 L	38.4 L	29.77x	30	6.71%
Influencer/Instagram	10.8 L	22.0 L	2.03x	427	2.60%

Email and SEO/Organic — the two “owned” channels — are dramatically more efficient than every paid channel, despite receiving a fraction of the total budget. Influencer/Instagram is the weakest performer: it has the highest CAC (Rs 427) and lowest ROAS, despite being the third-largest budget line.

3. Visual Analysis — Spend vs Revenue by Channel

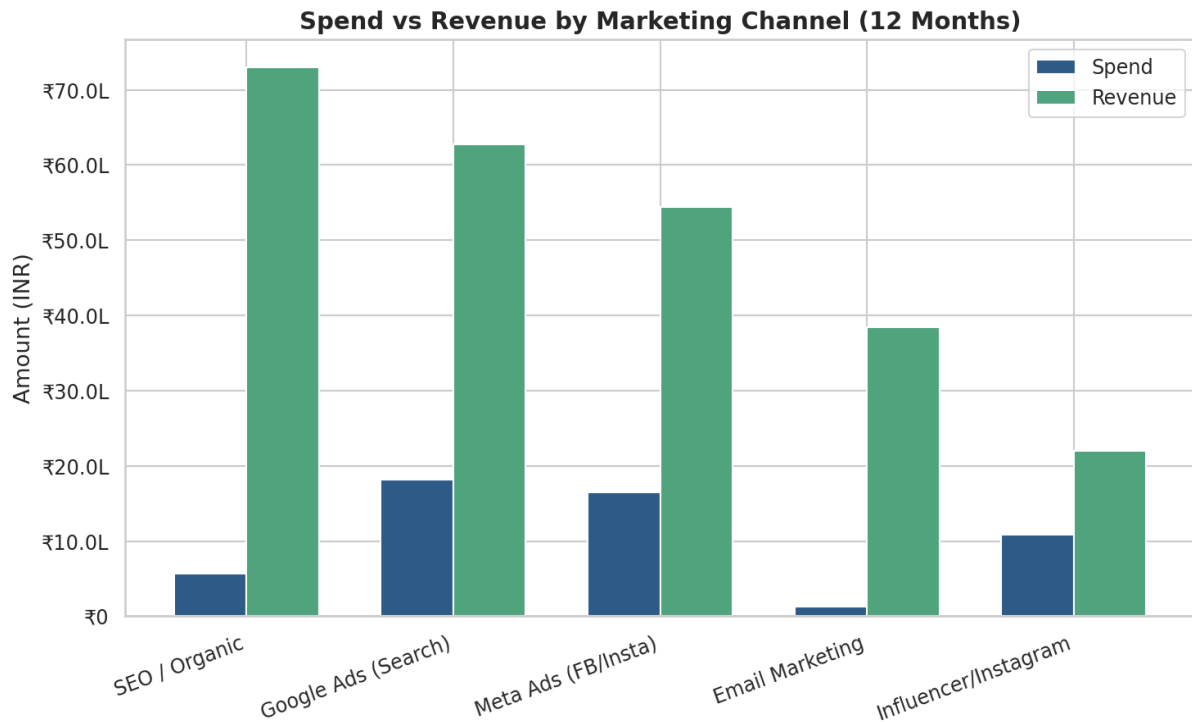


Figure 1: Total 12-month spend and revenue for each of the 5 marketing channels.

Google Ads and Meta Ads receive the largest share of budget, yet SEO/Organic — with a fraction of that spend — generates the highest total revenue of any single channel, a first signal that budget allocation and channel efficiency are currently misaligned.

4. Visual Analysis — ROAS by Channel

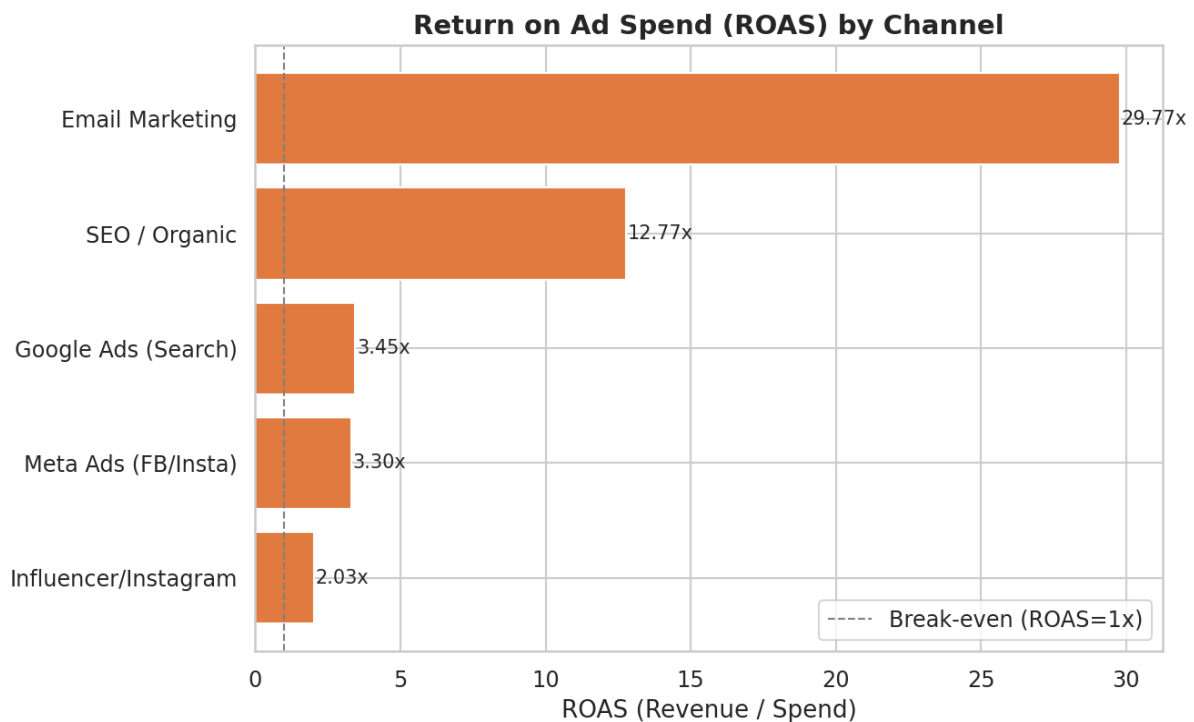


Figure 2: Return on Ad Spend (Revenue / Spend) by channel, with the 1x break-even line marked.

This is the clearest efficiency signal in the dataset. Email Marketing (29.8x) and SEO/Organic (12.8x) post returns far above the paid channels (2.0x-3.5x). This is a very common real-world pattern: owned channels are cheap to run and monetize an already-warm audience, while paid acquisition — especially influencer marketing — is expensive and harder to attribute directly to sales.

5. Visual Analysis — Monthly Seasonality Trend

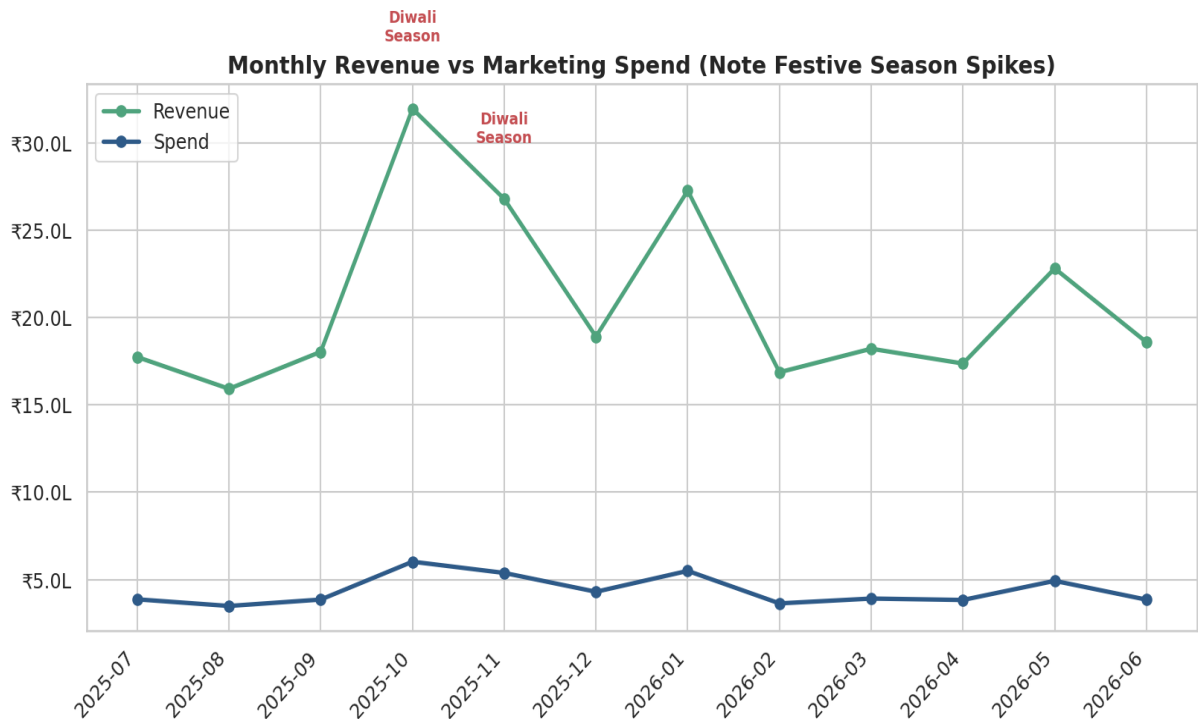


Figure 3: Monthly revenue and spend, showing festive-season demand spikes.

Revenue spikes sharply during the Diwali period (October-November) and again during the January Republic Day/New Year sale window, while spend stays comparatively flat across the year. This means ROAS is naturally highest exactly when seasonal demand is highest — suggesting the brand should **front-load** marketing budget slightly ahead of these windows to capture even more of the demand spike, rather than spreading spend evenly across the year.

6. Visual Analysis — Channel Efficiency Map

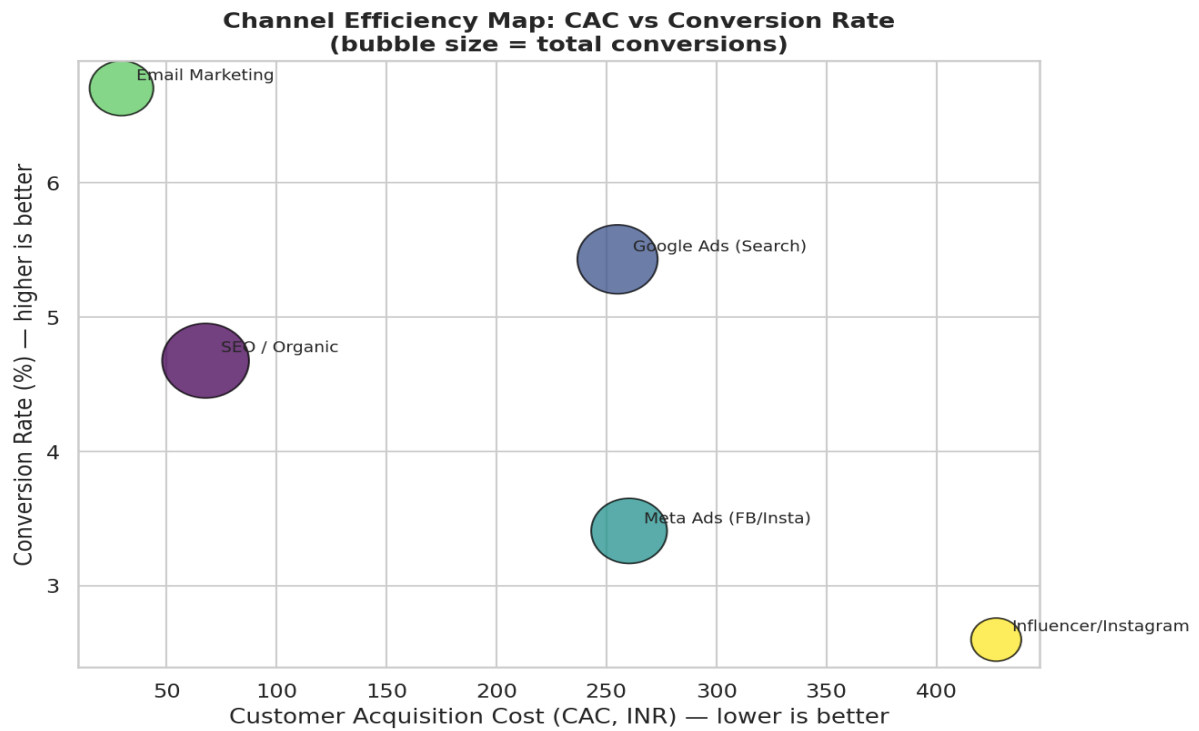


Figure 4: Customer Acquisition Cost vs Conversion Rate by channel (bubble size = total conversions).

The ideal channel sits toward the top-left of this chart: low CAC and high conversion rate. Email Marketing and SEO/Organic cluster there, while Influencer/Instagram sits in the least efficient corner — high cost per customer paired with the lowest conversion rate of any channel.

7. Customer Segment Analysis

Understanding who is actually buying helps target creative, offers, and regional strategy.

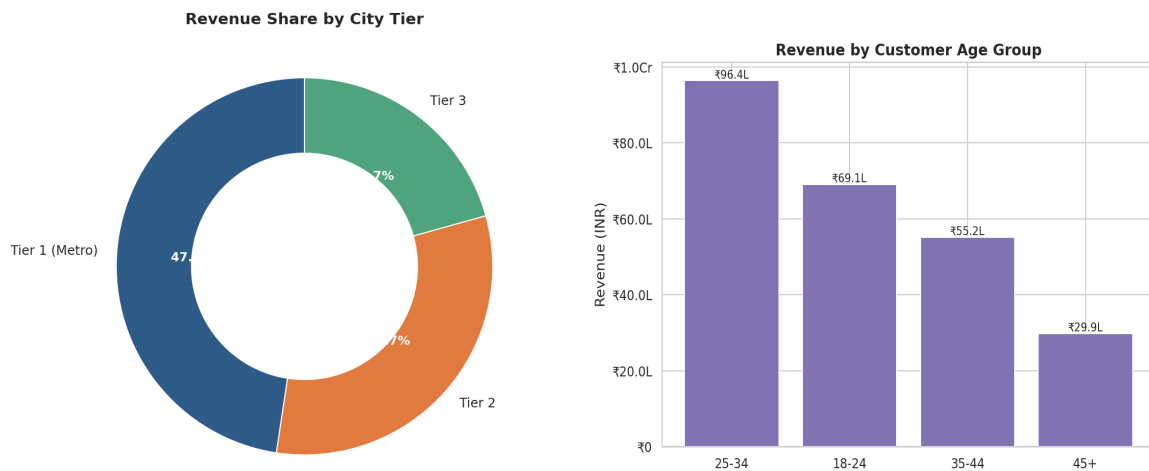


Figure 5 & 6: Revenue share by city tier (left) and by customer age group (right).

Tier-1 metro cities drive the largest single share of revenue, but Tier-2 and Tier-3 cities **combined** already make up roughly 40% of revenue — a segment that is easy to under-invest in if planning only follows where the “obvious” customer is. The 25-34 age group is the single largest revenue driver, consistent with a young, urban, wellness-focused buyer — useful context for creative and audience targeting decisions.

8. Marketing Funnel Overview

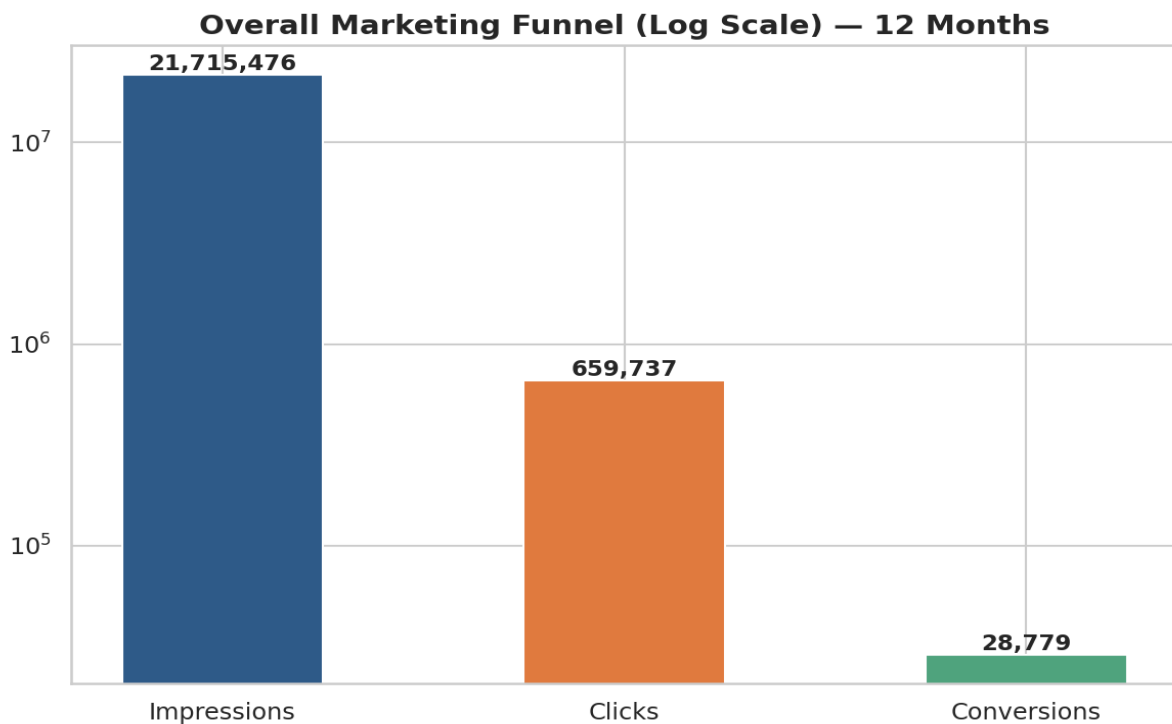


Figure 7: Overall Impressions -> Clicks -> Conversions funnel across all channels (log scale).

9. Key Findings & Business Recommendations

- **Owned channels outperform paid channels by a wide margin.** Email (29.8x ROAS) and SEO/Organic (12.8x ROAS) are dramatically more efficient than Google Ads, Meta Ads, or Influencer marketing, despite receiving a small fraction of total budget.
- **Influencer marketing is the weakest link.** It has the highest CAC (Rs 427) and lowest ROAS (2.03x) of any channel, despite being the third-largest spend line — a strong candidate for budget reduction or a stricter performance bar.
- **Seasonality is a growth lever, not just a pattern to observe.** Revenue peaks sharply around Diwali and the January sale window while spend stays flat, meaning the brand is likely leaving demand on the table during its highest-intent periods.
- **Tier-2/3 cities are a bigger opportunity than they look.** They already contribute roughly 40% of revenue with comparatively little dedicated spend or localized creative aimed at them.

Recommendations

- **Reallocate 15-20% of Influencer budget into Email and SEO/Content.** Influencer spend is currently the least efficient rupee spent, while email and organic are proven and clearly under-funded relative to their return.
- **Front-load ad spend 2-3 weeks before Diwali and the January sale window** to capture more of the seasonal demand spike while ROAS is naturally elevated.
- **Build a dedicated Tier-2/3 city strategy** — regional-language creative and micro-influencer partnerships — given this segment is already a large and growing share of revenue.
- **Reframe Google Ads and Meta Ads as reach engines, not efficiency engines.** Keep them for top-of-funnel reach and new-customer acquisition, but do not expect email-level ROAS from paid social or search.

"I built an end-to-end marketing analytics project simulating a year of campaign data for a D2C brand across 5 channels. I calculated the core KPIs — CTR, CVR, CAC, and ROAS — at the channel and monthly level, and found that email and organic search were delivering 3-10x better ROAS than paid channels despite getting a much smaller share of budget. I used that to build a concrete budget-reallocation recommendation, including accounting for festive seasonality like Diwali, which is something a lot of marketing reporting misses."

^ How this project can be described in an interview.

10. How This Project Was Built

Data: A synthetic but realistically-modeled dataset (no real company's data was used) — 1,825 daily rows across 5 channels and 12 months, with seasonal multipliers for Diwali, the January sale, and monsoon dips, plus randomized noise so figures resemble genuine campaign exports rather than a clean textbook dataset.

Analysis: Python (pandas, numpy) for KPI engineering — CTR, CVR, CPC, CAC, ROAS, ROI — aggregated at the channel, monthly, and customer-segment level. Visualizations built with matplotlib and seaborn, styled consistently for a professional, presentation-ready look.

Deliverables: A fully executed Jupyter notebook (code, charts, and narrative together), standalone chart exports, summary CSVs, and this PDF case-study report — mirroring the kind of documentation a real analytics team would produce for a quarterly business review.

This is a self-initiated portfolio project using a synthetic but realistic dataset modeled on real Indian D2C marketing benchmarks, built to demonstrate marketing analytics, KPI design, and data storytelling skills.